SOC 690S Machine Learning in Causal Inference
Midterm Review

This review highlights key concepts and proofs that you may pay particular attention to, as they may appear in various forms on the midterm exam. The scope of the exam is limited to the materials covered in the lecture notes and coding practices.

The exam will last 115 minutes (3:05PM-5:00PM), and you are allowed to finish and leave early. You are allowed to bring any materials, including lecture notes, textbooks, and external sources you print out. Laptops are *not* allowed. Calculators are not needed, but allowed. The exam will consist of two general parts.

- 12 multiple-choice questions; some are straightforward *yes/no/not certain* questions, while others require brief calculations
- 3 extended questions similar to the assignments
- In addition, there will be one bonus question on demonstrating *Neyman Orthogonality* for an identification method discussed in lecture, but for which the orthogonality property was not explicitly derived

Week 1: Motivation and Linear Regression

- Motivation for integrating machine learning into regression and causal inference
- Derivation of the *Frisch–Waugh–Lovell Theorem*
- The general concept of *adaptive statistical inference* and its implications for inference in Double Machine Learning

Week 2: Basic Machine Learning and Penalized Regression

- The rationale and properties of the *penalty term* in penalized regressions, including LASSO and Ridge
- The applicability of different penalized regression methods under various data-generating processes
- Tuning the penalty level in regularization using cross-validation and the plug-in approach

Week 3: Advanced Machine Learning Beyond Linear Models

- Proof of the bias–variance tradeoff and its role in tuning machine learning models
- Mechanisms of tree-based methods, including prediction rules in boosted trees
- How neural networks transform inputs into output predictions
- The rationale behind *gradient descent* and *backpropagation*
- How neural networks address overfitting

Week 4: Neyman Orthogonality and the Potential Outcome Framework

- Fundamental concepts in Double Machine Learning, including the *normal equation*, *score function*, and *moment condition*
- The procedure of *cross-fitting* and its role in statistical inference within Double Machine Learning
- Estimating treatment effects in Randomized Controlled Trials
- The concept of conditional ignorability and the proof of the *Horvitz–Thompson Theorem*

Week 5: Causal Inference Using Directed Acyclic Graphs (DAGs)

- The *backdoor criterion* and its equivalence to conditional ignorability
- Collider bias and a simple proof of statistical dependence when conditioning on a collider
- Basic understanding of the *front-door criterion* in causal inference

Week 6: Matching, Propensity Scores, and Weighting

- The relationship between matching and regression
- Derivation of the IPW estimand and its extensions for estimating ATT and ATC
- Understanding *common support* and *covariate balance* in propensity score estimation, and linking these concepts to machine learning (e.g., how ML overfitting can worsen *common support*; see Week 6 coding practice)
- Understanding Stabilized IPW and Augmented IPW estimators

Week 7: Instrumental Variable (IV) Estimation

- Understanding Omitted Variable Bias (OVB) and how instrumental variables address it
- Understanding IV estimation (the reduced form) and its properties, including (un)biasedness
- Understanding the four types of units (e.g., compliers, defiers) in IV estimation and their connection to the *Local Average Treatment Effect (LATE)*
- Understanding how IV estimation is implemented within Double Machine Learning